

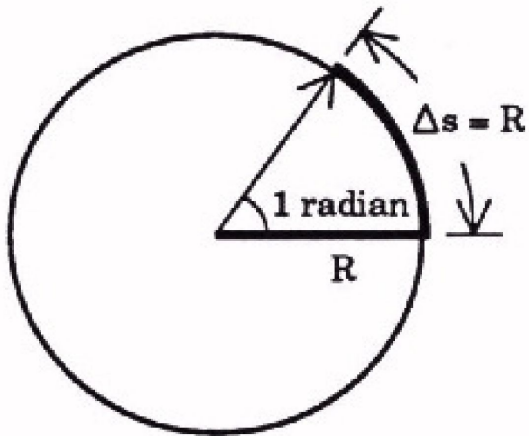
ROTATIONAL MOTION I

The radian

There are 2 types of pure unmixed motion:

- **Translational** - linear motion
- **Rotational** - motion involving a rotation or revolution around a fixed chosen axis(*an axis which does not move*).

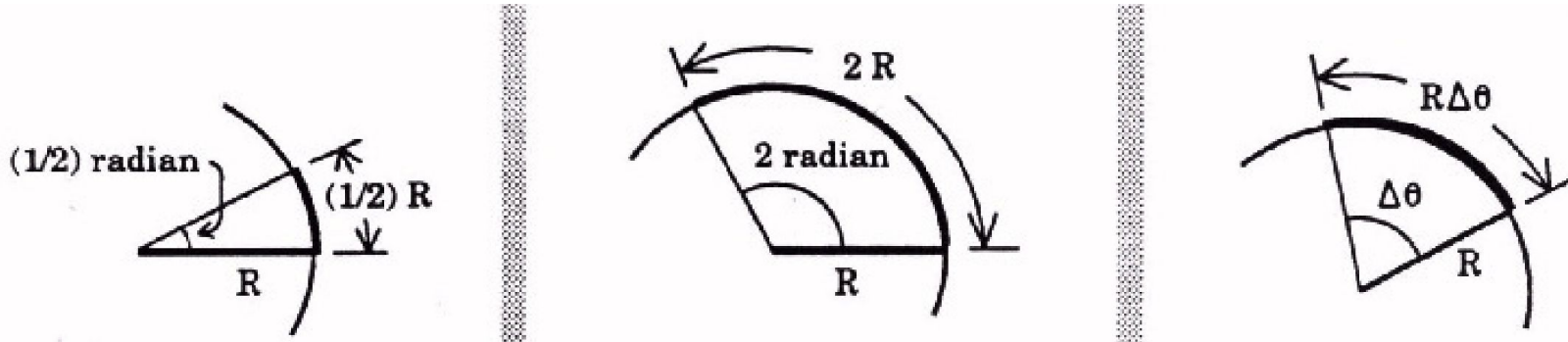
We need a system that defines **BOTH** types of motion working together on a system. Rotational quantities are usually defined with units involving a **radian** measure.



If we take the radius of a circle and **LAY IT DOWN** on the circumference, it will create an angle whose arc length is equal to **R**.

In other words, one radian angle is subtends an arc length Δs equal to the radius of the circle (**R**)

The radian



Half a radian would subtend an arc length equal to half the radius and 2 radians would subtend an arc length equal to two times the radius.

A general Radian Angle ($\Delta\theta$) subtends an arc length (Δs) equal to R .
The theta in this case represents **ANGULAR DISPLACEMENT**.

$$\Delta s = R\Delta\theta$$

Angular Velocity

Since velocity is defined as the rate of change of displacement. **ANGULAR VELOCITY** is defined as the rate of change of **ANGULAR DISPLACEMENT**.

$$\bar{v} = \frac{\Delta x}{\Delta t} \rightarrow \text{translational velocity}$$

$$\bar{\omega} = \frac{\Delta \theta}{\Delta t} \rightarrow \text{rotational velocity}$$

$$v = \frac{dx}{dt}, \omega = \frac{d\theta}{dt} \quad 1 \text{ revolution} = 2\pi \text{ radians} = 360^\circ$$

Angular Acceleration

$$\bar{a} = \frac{\Delta v}{\Delta t} \rightarrow \text{translational acceleration}$$

$$\bar{\alpha} = \frac{\Delta \omega}{\Delta t} \rightarrow \text{rotational acceleration}$$

Once again, following the same lines of logic. Since acceleration is defined as the rate of change of velocity. We can say the **ANGULAR ACCELERATION** is defined as the rate of change of the angular velocity.

Combining motions –Tangential velocity

$$\Delta s = R\Delta\theta$$

$$\frac{\Delta s}{\Delta t} = R \frac{\Delta\theta}{\Delta t}$$

$$v_T = \frac{\Delta s}{\Delta t}$$

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$$v_T = r\omega$$

First we take our equation for the radian measure and divide **BOTH** sides by a change in time.

The left side is simply the equation for **LINEAR** velocity. **BUT** in this case the velocity is **TANGENT** to the circle (*according to Newton's first law*). Therefore we call it **TANGENTIAL VELOCITY**.

Inspecting the right side we discover the formula for **ANGULAR VELOCITY**.

Therefore, substituting the appropriate symbols we have a formula that relates Translational velocity to Rotational velocity.

Tangential acceleration and rotational kinematics

$$v_t = r\omega \rightarrow \frac{v_t}{\Delta t} = \frac{r\omega}{\Delta t}$$

Using the same kind of mathematical reasoning we can also define Linear **tangential acceleration**.

$$a_t = r\alpha$$

*Inspecting each equation we discover that there is a **DIRECT relationship** between the Translational quantities and the Rotational quantities.*

We can therefore **RE-WRITE** each translational kinematic equation and turn it into a rotational kinematic equation.

$$v = v_o + at \rightarrow \omega = \omega_o + \alpha t$$

$$x = x_o + v_o t + \frac{1}{2}at^2 \rightarrow \theta = \theta_o + \omega_o t + \frac{1}{2}\alpha t^2$$

$$v^2 = v_o^2 + 2a\Delta x \rightarrow \omega^2 = \omega_o^2 + 2\alpha\Delta\theta$$

Example

A turntable capable of angularly accelerating at 12 rad/s^2 needs to be given an initial angular velocity if it is to rotate through a net 400 radians in 6 seconds. What must its initial angular velocity be?

$$\alpha = 12 \text{ rad/s}^2$$

$$\Delta\theta = 400 \text{ rad}$$

$$t = 6 \text{ s}$$

$$\omega_o = ?$$

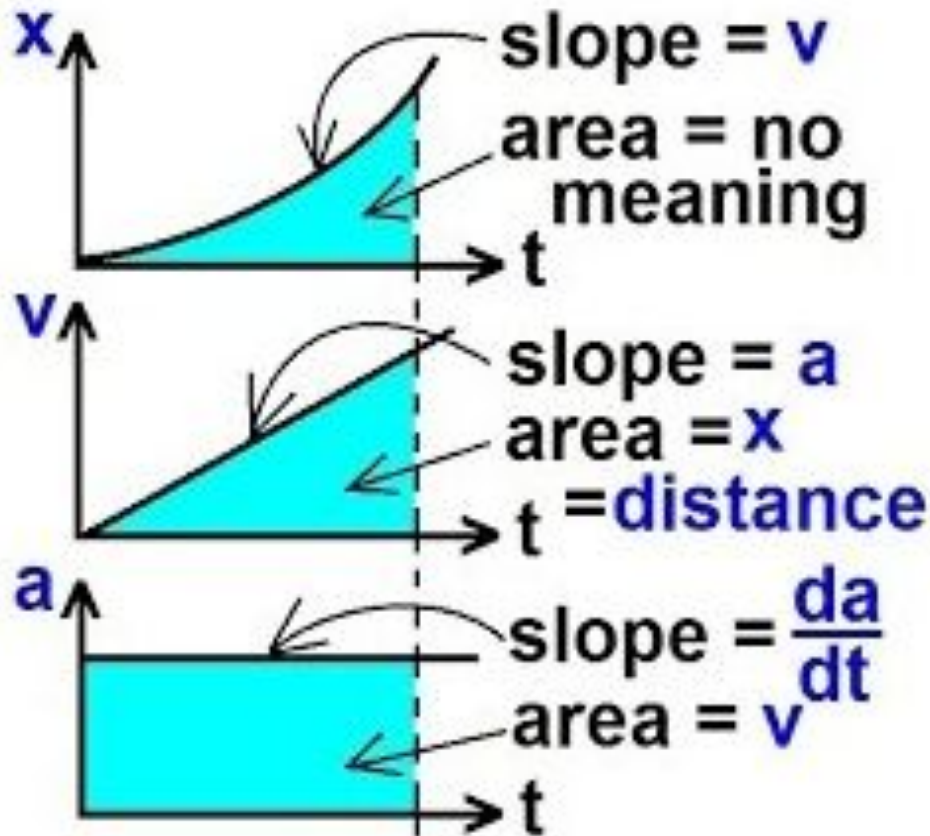
$$\Delta\theta = \omega_o t + \frac{1}{2} \alpha t^2$$

$$400 = \omega_o (6) + (0.5)(12)(6)^2$$

$$\omega_o = \textbf{30.7 rad/s}$$

MOTION GRAPHS

Linear Motion



Circular Motion

